Differing Priors and Information Acquisition When Others Make the Decision*

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Abstract

Should you pay more for information when someone else will act on it instead of you? Standard Bayesian theory predicts it should not matter, and this paper tests that prediction experimentally.

In a laboratory experiment, pairs face the Bayesian decision problem of purchasing a signal about a binary state. I elicit first- and second-order posterior beliefs to evaluate the effects of different priors on belief updating. Then, I independently vary (i) whether participants share common or differing prior beliefs when eliciting their posteriors, and (ii) whether the information purchaser is also the final decision maker.

Three main results emerge. First, participants expect others to update beliefs similarly to themselves. Second, willingness to pay (WTP) is significantly higher when participants implement their own strategy under differing priors, consistent with an amplified control premium when belief differences are salient. Third, individual deviations from Bayesian updating predict WTP in theoretically consistent directions.

These findings suggest that belief diversity can increase the perceived value of decision rights. This has implications for organizational information acquisition and delegation design.

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